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To the Editor,
Communications AI & Computing

Dear Editor,

We are pleased to submit our manuscript “OmniGene-4: A Unified Bio-Language MoE Model with Router-Level Interpretability” for consideration by *Communications AI & Computing*.

The paper addresses a question that has been actively debated for general LLMs but rarely measured directly in the biological domain: how do multi-modal large language models that jointly process natural language and biological sequences (DNA, protein, structural alphabets) actually answer biological questions, especially sequence-grounded questions whose answer depends on residue-level patterns rather than literature recall? We use the discrete router state of a Mixture-of-Experts (MoE) architecture as a direct observable to dissect this question.

We build OmniGene-4, a unified bio-language foundation model on Gemma-4-26B-A4B (128 experts per layer, top-8 routing), and train it through a five-stage pipeline (continued pretraining + four supervised fine-tuning rounds, including a final dual-head architecture with per-residue 3Di and DSSP classifiers).

Three findings:

(1) Router-level decomposition of CPT vs SFT. By installing forward hooks on every router across eight task families, we find that continued pretraining accounts for 96% of cross-task expert differentiation, while supervised fine-tuning accounts for 4%, with bootstrap 95% confidence intervals excluding zero for both. CPT reshapes middle transformer layers; SFT concentrates on the final two layers. To our knowledge this is the first router-level CPT/SFT decomposition for a biological MoE model.

(2) Gate-versus-expert separation. Within the protein-homology task family, per-pair routing divergence stays below 0.04 (versus 0.23 cross-task), implying that homology decisions occur *inside* expert computation rather than at the routing gate. The gate selects the modality; the experts compute the answer.

(3) Strong benchmark performance. On a balanced protein remote-homology evaluation, the model reaches 82.60% accuracy, outperforming ESM-2 (3B), MMseqs2, and DIAMOND by 28–31 percentage points on the identical pair set. Standard homology reaches 99.40%, and BixBench (a general biological-knowledge benchmark covering literature comprehension, mutation interpretation, and bioinformatics workflow reasoning) reaches 93.66%. The dual-head 3Di and DSSP classifiers reach 78.6% and 100% per-residue accuracy, without interfering with chat-style generation.

All six trained model variants and the complete codebase are publicly released under github.com/maris205/omnigen and huggingface.co/dnagpt. A preliminary version is available on bioRxiv (DOI: 10.1101/2026.01.03.697478).

We believe the work is well suited to *Communications AI & Computing* because it combines a

methodological contribution (the first router-level interpretability decomposition for a multi-modal bio-foundation MoE) with a concrete application (sequence-grounded biological reasoning), and produces fully reproducible artefacts. The manuscript has not been submitted elsewhere.

Thank you for considering our work.

Sincerely,

Liang Wang